

Problem 4

A body with mass $m = 10$ kg is transferred from Earth to the Moon. Calculate:

- (a) The gravitational force acting on the body on Earth and on the Moon.
- (b) The force shown by a stationary dynamometer if the body is hung from it on Earth and on the Moon.
- (c) The weight of the body when placed on the ground of both celestial bodies.
- (d) The weight of the body in a rocket during launch from Earth, at the moment when the rocket is accelerating vertically upward with acceleration $a = 4g$.
- (e) The weight of the body in a rocket during launch from the Moon, under the same acceleration conditions as on Earth.

The acceleration due to gravity on Earth is $g_z = 9.81$ m/s², and on the Moon is $g_M = 1.62$ m/s².

Solution

(a) Gravitational force on Earth and Moon

The gravitational force is given by:

$$F_g = mg.$$

On Earth:

$$F_{g,z} = 10 \times 9.81 = 98.1 \text{ N.}$$

On the Moon:

$$F_{g,M} = 10 \times 1.62 = 16.2 \text{ N.}$$

(b) Force shown by the dynamometer

Since a stationary dynamometer measures the gravitational force, the readings will be the same as the gravitational forces calculated above:

- On Earth: 98.1 N. - On the Moon: 16.2 N.

(c) Weight of the body on the ground

Weight is just the gravitational force, so:

- On Earth: 98.1 N. - On the Moon: 16.2 N.

(d) Weight of the body during Earth rocket launch

During launch, the apparent weight increases due to acceleration:

$$W = m(g_z + a).$$

With $a = 4g_z$:

$$W = 10(9.81 + 4 \times 9.81).$$

Calculate:

$$W = 10(9.81 + 39.24) = 10(49.05) = 490.5 \text{ N.}$$

(e) Weight of the body during Moon rocket launch

For the Moon, under the same conditions, with $a = 4g_M$:

$$W = m(g_M + 4g_M).$$

Simplify:

$$W = 10(1.62 + 4 \times 1.62).$$

Calculate:

$$W = 10(1.62 + 6.48) = 10(8.1) = 81 \text{ N.}$$

Final Answer:

- Gravitational force on Earth: 98.1 N. - Gravitational force on the Moon: 16.2 N. - Dynamometer reading on Earth: 98.1 N. - Dynamometer reading on the Moon: 16.2 N. - Weight on Earth: 98.1 N. - Weight on the Moon: 16.2 N. - Weight during rocket launch from Earth: 490.5 N. - Weight during rocket launch from Moon: 81 N.